Shivam Maurya

Robotics Software Engineer | ROS 2, Motion Planning, Simulation and Autonomous Systems

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SUMMARY

Robotics Software Engineer (B.E., CGPA 8.06 / 10) with extensive hands-on experience in ROS 2 (Humble, Jazzy), MoveIt 2, Nav2, and NVIDIA Isaac Sim. Core contributor to the ROS 2 ecosystem with 15+ merged pull requests across navigation2 and ros2_control. Experienced in multi-robot task orchestration, digital twin physics tuning, URDF/Xacro kinematic modeling, real-time hardware interfaces, and sim-to-real perception deployment. Built a 30-robot simulation at 40-50 FPS, a Nav2 autonomous navigation stack, and a 1 kHz ros2_control interface for STM32. All India Rank under 20, ABU Robocon.

TECHNICAL SKILLS

Robotics Frameworks: ROS 2 (Humble, Jazzy, Kilted), rclcpp, rclpy, Nav2, MoveIt 2, ros2_control, ros2_controllers, hardware_interface, TF2, URDF, Xacro, Behavior Trees, OMPL, Lifecycle Nodes

Simulation and Digital Twins: NVIDIA Isaac Sim, Omniverse, USD / USDA Instancing, PhysX Tuning, Gazebo (Classic, Harmonic), RViz2, Sim-to-Real Transfer, Synthetic Data Generation

Navigation, Planning and Control: SLAM, AMCL Localization, DWB Local Planner, Costmaps, Motion Planning, Trajectory Generation, Kinematics and Dynamics, PID Control, State Estimation (EKF), Sensor Fusion, 2D LiDAR, Real-Time Control

Languages and Build: C++ (C++14 / C++17), Python 3, Bash, CMake, colcon, YAML, SQL

Embedded and Middleware: STM32 (F4, H7), Arduino, ESP32, Micro-ROS, UART, DMA, FastDDS (UDP tuning), Custom Binary Protocols (CRC8), Zenoh-Pico

Perception and Edge AI: OpenCV, YOLOv8, U-Net, ONNX Runtime, TensorRT, PyTorch, TensorFlow, Model Benchmarking, CPU-Only Inference Optimization

DevOps and Tools: Docker, Git, GitHub, Linux (Ubuntu 22.04 / 24.04), CI/CD, pytest, ament_lint, ament_mypy, Flask, FastAPI

PROJECTS

Multi-Arm UR10e Pick-and-Place with NVIDIA Isaac Sim | ROS 2 Jazzy, MoveIt 2, Isaac Sim, Gazebo Harmonic, PhysX

- Built a production-grade 3-arm UR10e and Robotiq 2F-140 relay workcell (stations A to D) with custom PickPlace action interfaces, an S-curve trajectory solver, and a sequential state-machine task manager, running identically in RViz, Isaac Sim, and Gazebo.
- Developed a procedural multi-cell spawner creating 10 tri-arm groups (30 UR10e robots, 20 RTX pinhole cameras) at 40-50 FPS on a single consumer GPU (RTX 5060 Ti 16GB); 3 groups (9 robots) reach 65.25 FPS.
- Integrated a tactile sensing array (PhysxContactReportAPI across 7 rigid bodies per robot) and a calibrated camera array for physical AI foundation model dataset generation.
- Applied critically damped position drives (Kp=5000, Kd=1000) and frame-0 joint initialization for stable, zero-jitter GPU physics simulation.

Autonomous Mobile Robot Navigation with ROS 2 and Nav2 | ROS 2 Humble, Nav2, AMCL, SLAM Toolbox, Gazebo | github.com/ShivamMaurya14/rosnav

- Implemented a decoupled 3-tier Nav2 pipeline with a lifecycle-managed map server (unconfigured to active), KLD-sampled AMCL localization (500-2000 particles), and global and local planning with custom recovery behaviors.
- Configured a DWB local trajectory planner with 7 weighted scoring critics (path alignment, goal alignment, obstacle clearance, oscillation) and a 3-tier costmap.
- Built a Gazebo Classic and RViz2 simulation with a custom URDF/Xacro differential-drive robot, 360-degree 2D LiDAR, and occupancy-grid maps for autonomous goal navigation.

ROS 2 Dual-Mode Motor Interface for STM32 and Arduino | ros2_control, STM32 (F4/H7), Micro-ROS, C++ | github.com/ShivamMaurya14/ros2_control_motor_interface

- Developed a ros2_control SystemInterface plugin with dual-mode communication: a custom CRC8-checked binary UART protocol (over 1 kHz deterministic loops) and Micro-ROS native DDS, exposing position, velocity, and effort interfaces.
- Designed hardware-abstracted firmware templates for STM32 (F4/H7) and Arduino with DMA-enabled UART, 1 kHz hardware timer interrupts, encoder mode, and real-time PID loops.
- Added a mock-hardware toggle so the same launch configuration runs on physical hardware or in simulation with no code changes.

RealFake Classifier: Real-Time Perception Pipeline for ABU Robocon | ONNX Runtime, YOLOv8, ROS 2, Flask | github.com/ShivamMaurya14/RealFake-Classifier-Robocon

- Engineered a CPU-only ROS 2 perception pipeline with hot-swappable ONNX, YOLOv8, and TensorFlow backends; ONNX Runtime reaches 3.89 ms (257 FPS) against 22.18 ms for TensorFlow and about 18.5 ms for YOLOv8.
- Built a robot decision engine mapping bounding boxes to gripper commands (PICK_REAL, REJECT_FAKE) and alignment targets (center offset dx/dy, heading error), plus a Flask dashboard with MJPEG streaming and live telemetry.

EXPERIENCE

Physical AI Intern | Nextup Robotics | Remote | Aug 2026 - Oct 2026

- Tuned FastDDS non-blocking UDP transports for zero-packet-drop synchronization between MoveIt 2 and Isaac Sim OmniGraph bridges (/clock, joint states).
- Orchestrated a multi-step 3-arm workflow across four stations with custom ROS 2 action servers and state machines, and scaled the simulation to 30 UR10e robots at 40-50 FPS on one consumer GPU.
- Integrated tactile contact sensing (7 rigid bodies per robot) and 20 calibrated RTX cameras to generate vision and tactile datasets for NVIDIA Cosmos foundation models.
- Authored 7 engineering guides covering workcell setup, calibration, and headless CI deployment.

Software Lead | Centre of Excellence for AI & Robotics, Army Institute of Technology, Pune | Jun 2024 - Present

- Led software design and hardware-software integration for the ABU Robocon national robot (All India Rank under 20), deploying onboard real-time CNN perception directly into navigation decision loops.
- Built an ONNX export pipeline and a hot-swappable inference backend (30+ FPS on CPU-only hardware) enabling runtime model swaps and A/B testing with zero downstream code changes.
- Directed a 100+ member research cell and organized hackathons, workshops, and project sprints across 5+ teams.

Open-Source Contributor | ROS 2 Ecosystem (navigation2, ros2_control, ros2_controllers, ros2_control_demos, ros2/demos) | Jun 2025 - Present

- Authored 15+ merged pull requests (5+ more under review) in foundational robotics libraries, working with global maintainers through code review.
- Documented real-time safety patterns and thread execution constraints in ros2_control controller_interface headers to protect deterministic actuator loop timing.
- Diagnosed runtime crashes in gz_ros2_control by analyzing backtraces through ResourceManager and ControllerManager initialization checks, and identified the failure boundary.
- Stabilized CI: fixed cross-repository build failures, migrated ros2_control spawner configurations for the ROS 2 Kilted release, resolved headless display errors, and introduced ament_mypy static type checking.

EDUCATION

B.E., Automation and Robotics Engineering | Army Institute of Technology, Pune, India | Jun 2024 - May 2028 (Expected) | **CGPA: 8.06 / 10**

Coursework: Robotics Systems, Kinematics and Dynamics, Control Theory, Signals and Systems, State Estimation, Sensor Fusion, Machine Learning, Computer Vision, Data Structures and Algorithms, Object-Oriented Programming

HONORS & ACHIEVEMENTS

- All India Rank under 20, ABU Robocon 2025/26 (Lead Software Architect)
- Top 500 All India, Amazon ML Challenge 2026 (F0.5 score 0.9602)
- Runner-Up, Hackathon 2026 (CREDI-MITRA) | Top Submission, CODECURE Hackathon 2025, IIT BHU
- State Level Round 2, Avishkar Maharashtra Inter-University Research Convention 2026 | Finalist, IIIT Pune Thinking Machines 2025

Languages: English (Professional Working Proficiency), Hindi (Native)